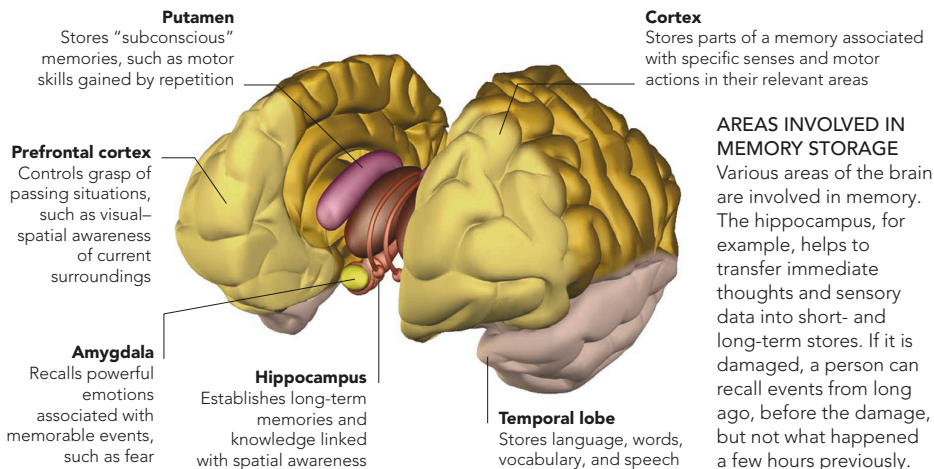


MEMORY AND RECALL

Memories are the brain's information store. No single region of the brain processes them as they are being established, nor acts as a storage site for all memories.

These processes depend on the significance and time span of the memory, its depth of emotional impact, and its association with specific senses such as eyesight.



THOUGHT IN ACTION

The real-time scanning method fMRI (functional magnetic resonance imaging) reveals tiny localized increases in blood flow. As a result, the scans can pinpoint

which areas of the brain are busy during well-defined mental activities, such as studying the visual details of an image, or listening to and understanding speech.

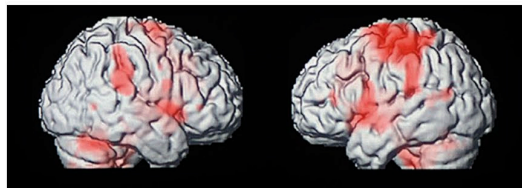


RIGHT SIDE OF BRAIN

LEFT SIDE OF BRAIN

PLANNING A MOVEMENT

The subject of this fMRI scan was asked to think about performing a task during the scan. The image shows activity in both the left and right prefrontal areas and also in both the left and right auditory cortex.



RIGHT SIDE OF BRAIN

LEFT SIDE OF BRAIN

MAKING THAT MOVEMENT

When actually performing the task, large parts of the premotor and motor cortex show up on the brain's left side. The cerebellum (at the base of the brain) helps to control precise muscle coordination.